

Hotel Demand Forecasting and AI Planning

From Revenue Intelligence to Agentic Operations

AI in Business and Data Monetization

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Business Framing

The Revenue Director asks:

1

Which hotels are at risk, and which are our growth drivers?

2

Can we forecast weekly revenue for Q1 2021 to size staffing and procurement?

3

How should THEHOTEL use AI end-to-end, from insight to action?

Who uses this

Revenue Director
Finance, Operations, Procurement

Decision supported

Q1 2021 staffing and procurement sizing by hotel and by week

Planning horizon

13 weeks
Weekly granularity

Cost of error

Asymmetric.
Over-forecast: recoverable idle cost.
Under-forecast: lost revenue, understaffed service, reputational risk.

For growth hotels, under-forecasting is not recoverable. For hotels still rebuilding post-COVID, over-commitment to headcount is the greater risk. Forecasts must be interpreted by segment.

EDA: Four Key Findings

1 Revenue scale: 4.1x gap across hotels

Hotel 29 averages \$1.62M per week. Hotel 59 averages \$0.39M. Any group-level forecast masks hotel-specific planning needs entirely.

2 Three distinct hotel segments

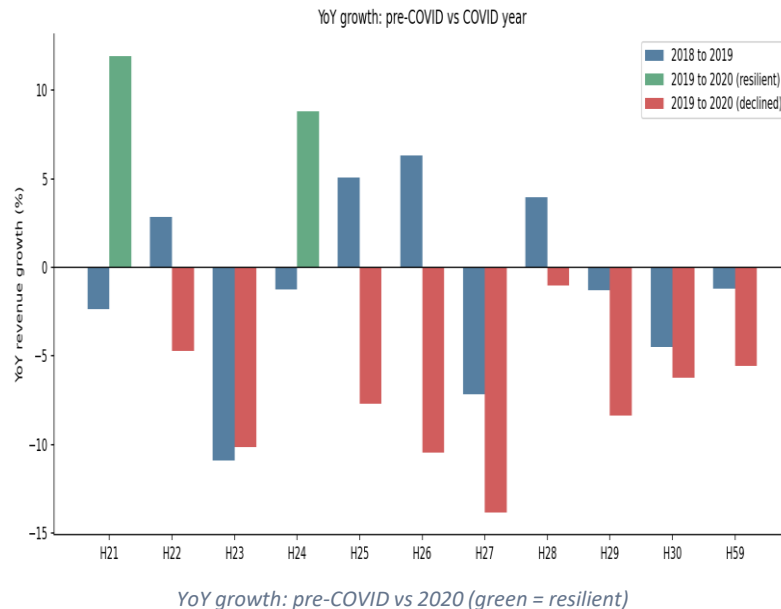
Growth drivers (Hotels 22, 25, 26, 28, 29): CAGR* above 3% pre-COVID. Declining (Hotels 23, 24, 27): negative CAGR. Stable (Hotels 21, 30, 59): flat trend, low volatility.

3 COVID impact was uneven

Hotels 21 and 24 grew in 2020. Hotels 26 and 27 fell more than 10%. A single 2020 adjustment factor applied to all hotels would be misleading.

4 Seasonality: week 13 is the group peak

Week 13 is peak for 6 of 11 hotels. Hotels 23, 25, and 27 peak earlier (weeks 9 to 10). Procurement and staffing plans must follow hotel-specific patterns, not a group average.



* CAGR = compound annual growth rate, computed on 2016-2019 pre-COVID years only: $(2019 / 2016)^{(1/3)} - 1$.

Full figures: Appendix A1 (revenue scale), A2 (segmentation bubble), A3 (seasonality)

Feature Engineering and Modeling Dataset

Feature	Description
<code>prev_year_sales</code>	Sales from the prior year, same hotel and week
<code>rolling_mean_2yr</code>	Average of the last two years, same hotel and week
<code>hist_avg</code>	Cumulative mean of all prior years for that hotel and week
<code>yoy_growth</code>	Year-over-year momentum (prior year vs year before)
<code>week_seasonality_idx</code>	Hotel's weekly sales relative to its own annual average
<code>hotel_avg</code>	Hotel-level mean weekly revenue across training years
<code>segment_encoded</code>	Segment: 0 declining, 1 stable, 2 growth driver
<code>hotel_cv</code>	Year-to-year revenue volatility (coefficient of variation)
<code>week</code>	Week number 1 to 13

Raw CSV

11 hotels 5 years 13 weeks

Reshape

Wide to long format

Feature computation

Lag, rolling, seasonality

Train (2018)

143 rows 9 features

Validate (2019)

143 rows held out

Forecast input (2021)

143 rows no actuals

Forecasting Methodology: Baseline vs Ridge Regression

Baseline: historical average

What it does

For each hotel and week, predict the mean sales across all training years. No model learning, no features.

Why it is the benchmark

It captures hotel-specific and seasonal patterns directly. Any improved model must outperform this to justify its complexity.

Limitation

Ignores trend direction. A growth driver and a declining hotel are treated identically if their historical means are similar.

Improved: Ridge regression

Why Ridge over alternatives

With 143 training rows, regularisation prevents overfitting. Ridge handles correlated features (lag, rolling mean, historical average) without dropping any.

Alpha selection

5-fold cross-validation on the 2018 training set. Best alpha: 4.86. LOOCV was considered but rejected due to high variance from within-hotel autocorrelation.

Top predictors

Week seasonality index, hotel average revenue, and prior year sales carry the largest coefficients. See Appendix A5 for full coefficient chart.

Evaluation and FY2021 Q1 Forecast

Hotel	Segment	MAPE %	Model	Decision	Q1 2021 Rec.
H21	stable	8.8%	Ridge	Medium confidence	\$9.64M
H22	growth driver	5.6%	Ridge	High confidence	\$8.93M
H23	declining	9.3%	Ridge	Medium confidence	\$9.80M
H24	declining	4.0%	Ridge	High confidence	\$9.87M
H25	growth driver	3.6%	Ridge	High confidence	\$19.18M
H26	growth driver	3.5%	Ridge	High confidence	\$8.51M
H27	declining	10.0%	Ridge	Flag for review	\$11.02M
H28	growth driver	17.0%	Ridge	Low confidence	\$15.13M
H29	growth driver	4.2%	Baseline	Use baseline	\$22.50M
H30	stable	7.5%	Baseline	Use baseline	\$6.98M
H59	stable	4.7%	Ridge	High confidence	\$5.20M

\$126.8M

Group Q1 2021 forecast (+1.2% vs 2019 \$125.3M)

Appendix A4: per-hotel MAPE comparison chart.

Decision Rationale: Ridge is selected only where it improves on the historical mean baseline by more than 0.5 pp; confidence tier then reflects validation MAPE on the 2019 hold-out.

Business Recommendations

Q1: Which hotels are at risk, and which are growth drivers?

- Growth drivers (Hotels 22, 25, 26, 28, 29): pre-COVID CAGR above 3%. Prioritise capacity and staffing investment. High confidence forecasts available for 22, 25, 26.
- Structural risk (Hotels 23, 24, 27): negative pre-COVID CAGR. Hotels 23 and 27 show forecast recovery vs 2019, but this reflects the 2019 anchor, not a confirmed trend reversal. Monitor closely before committing resources.
- Hotel 28: growth driver segment with high forecast uncertainty (17% MAPE). Revenue Director should review manually before Q1 planning lock.

Q2: Can we forecast Q1 2021 to size staffing and procurement?

- Yes, with differentiated confidence. Five hotels at high confidence (MAPE below 6%): use Ridge forecasts directly. Two at medium confidence (Hotels 21, 23): apply a 10% planning buffer. Hotels 27 and 28: need human review. Hotels 29 and 30: use historical mean baseline.
- Group Q1 total: \$126.75M, up 1.2% vs 2019. Week 13 is peak for 6 hotels. Hotels 23 and 25 peak at week 10. Procurement timelines should reflect hotel-specific patterns, not a single group schedule.

Q3: How should THEHOTEL embed AI into planning?

- Weekly forecast refresh from updated actuals, automated alert when any hotel deviates more than 10% from plan. GenAI copilot surfaces root cause and suggests corrective action in plain language.
- Human approval required before any staffing or procurement change above a defined threshold. Agentic workflow automates low-risk adjustments; escalates high-impact decisions to the Revenue Director.
- Slides 7 and 8 specify the copilot design and agentic workflow.

GenAI Copilot: Quarterly Weekly Revenue Intelligence

System prompt

You are a revenue planning copilot for THEHOTEL Group. Translate forecast outputs into concise briefings for the Revenue Director. Use business language, no technical jargon. Ground every claim in model output. Flag low-confidence forecasts explicitly. Recommend actions that remain subject to human validation before any operational commitment.

Input context

- Weekly recommended forecast per hotel
- Q1 cumulative forecast vs 2019 actuals
- Model confidence tier per hotel
- Validation MAPE from 2019 hold-out
- Hotel segment: growth driver, stable, declining
- Peak week per hotel (from EDA)
- Group Q1 total and YoY change
- Business rule: 10% deviation triggers alert

Output format

Mode 1: Pre-Quarter Planning Brief

- Executive summary (group level)
- Top 3 risks and top 3 opportunities
- Hotels requiring manual review
- Operational actions: staffing, procurement, commercial
- Limitations note

Group coordination requires quarterly horizons for commitment decisions. Weekly alerts serve a different cadence: monitoring whether the plan holds as actuals flow.

Mode 2: Weekly Deviation Alert

- Actual vs forecast per hotel (week N)
- Deviation flags (threshold 10%)
- Trend direction and recommended response: hold, adjust, or escalate

Example output

Mode 1: Pre-Quarter Planning Brief

Q1 2021 group forecast: \$126.8M, up 1.2% vs 2019. Seven hotels stable or improved.

Top risk: Hotel 27 and 28 carries 10% and 17.0% forecast error. Q1 total \$11.02M and \$15.13M are directional only. Staffing and procurement require sign-off before order.

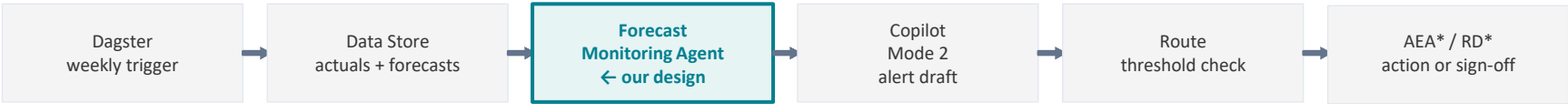
Top opportunity: Hotels 25 and 26 show highest model confidence (error below 3.6%). Hotel 25 peaks at week 10 at \$1.74M. Activate procurement by week 7.

Manual review: Hotel 27 (indifferent model), Hotel 28 (low confidence).

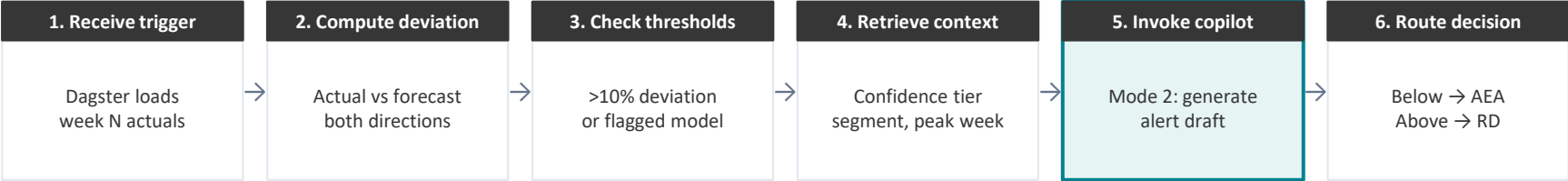
Mode 2: Weekly Deviation Alert

Week 3 alert: Hotel 23 actual \$612K vs forecast \$720K, deviation of 15%. Pattern consistent with weeks 1 and 2. Recommend escalation to Revenue Director before week 4 procurement confirmation.

Agentic Workflow: Forecast Alert to Action



Agent steps



* Action Exec. Agent
* Revenue Director

Human approval required when

- Financial impact above \$50K per hotel per week
- Hotel model decision is Low confidence, Indifferent, or Use baseline
- Staffing change above 10% of planned headcount
- Any commercial campaign activation or pause

Audit trail

- Week, hotel, actual vs forecast, deviation %
- Confidence tier and model version at trigger
- Copilot Mode 2 draft generated
- Routing decision and RD outcome if escalated

Errors and failures

- Failed action with error reason
- Actual outcome (for recalibration)

Failure modes and safeguards

What could go wrong	Safeguard
Agent fires on stale data	Dagster validates freshness before triggering agent
Escalation rate too high — RD flooded	Review monthly, recalibrate if above 3 per week
Copilot explanation unsupported by data	System prompt restricts to model outputs only
Action Execution Agent fails post-approval	Retry once, then re-escalate with error log

A0: Full System Pipeline and Design Rationale

Pre-quarter schedule (Dagster)



Weekly monitoring (Forecast Monitoring Agent)

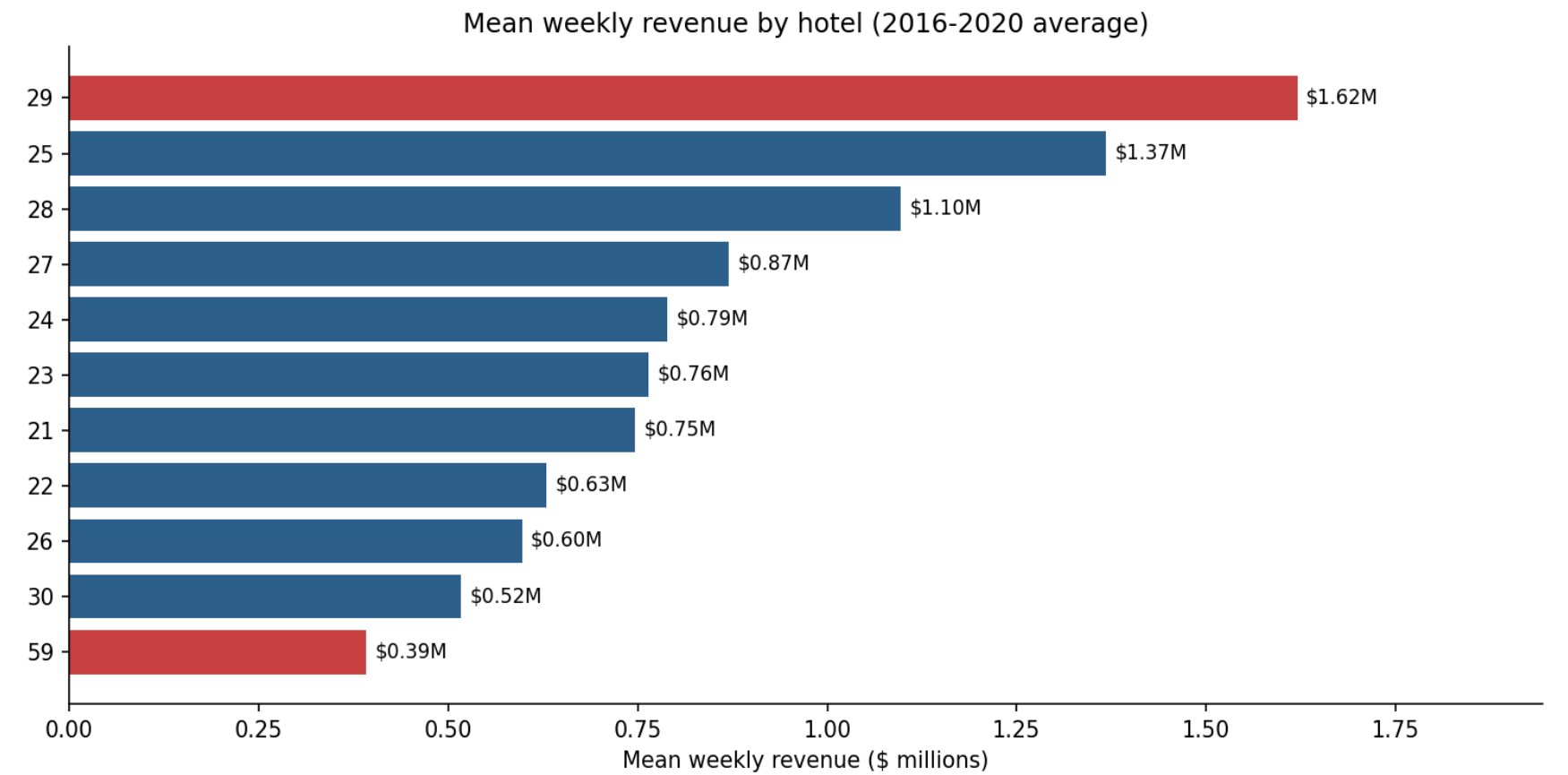


* out of scope

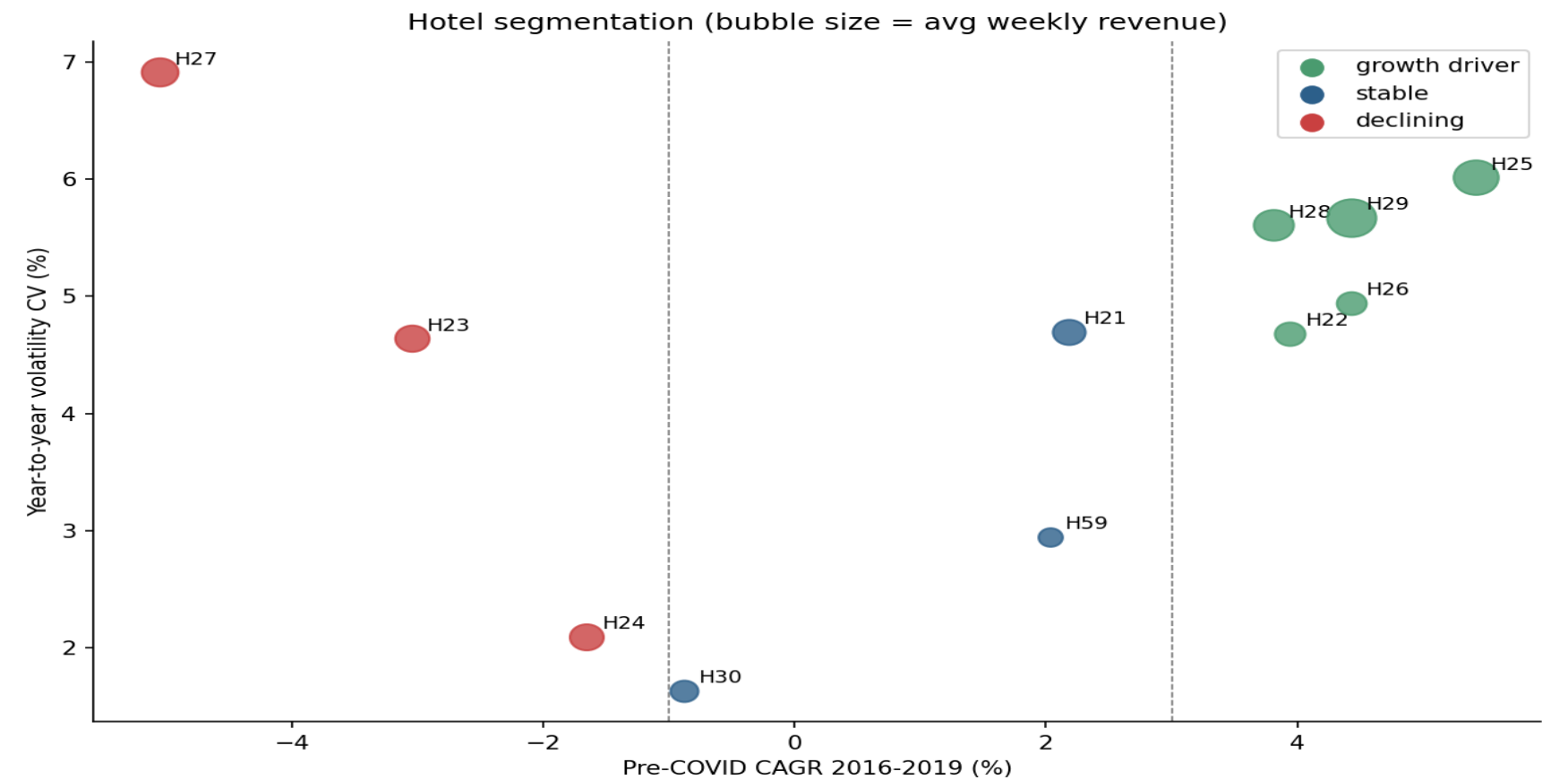
Design rationale

- Data collection is handled by Dagster: stable, deterministic, and significantly cheaper than agent token costs.
- The Action Execution Agent is acknowledged but out of scope; it requires integration with HR and procurement systems beyond the current data scope.
- The Forecast Monitoring Agent sits at the boundary between automated computation and human decision — the highest-value intervention point in the pipeline.
- **The copilot serves two distinct roles: Mode 1 informs commitment decisions before the quarter begins; Mode 2 informs escalation decisions when automated action is blocked above threshold.**

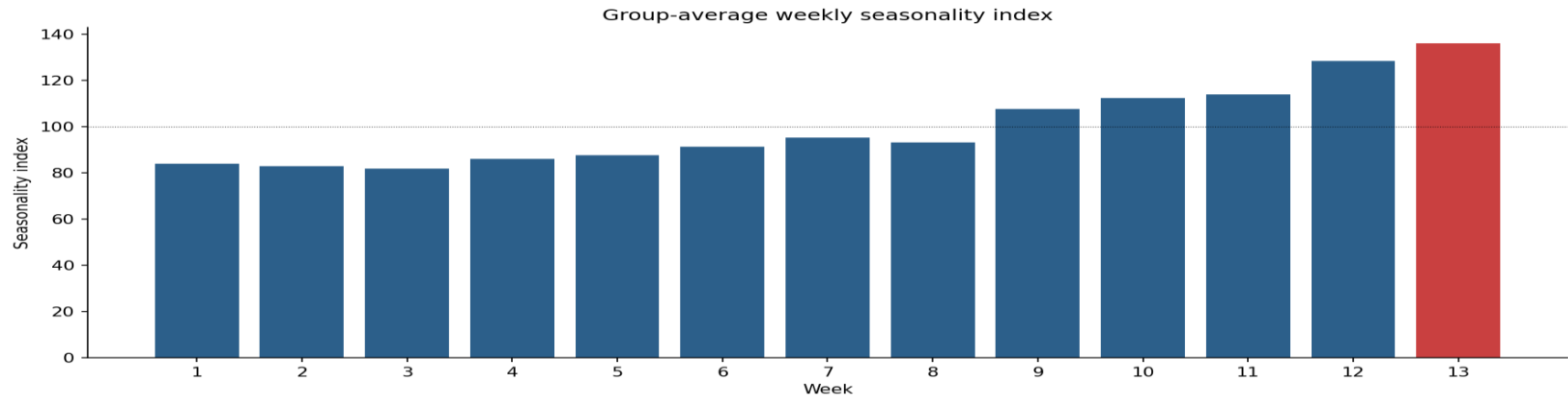
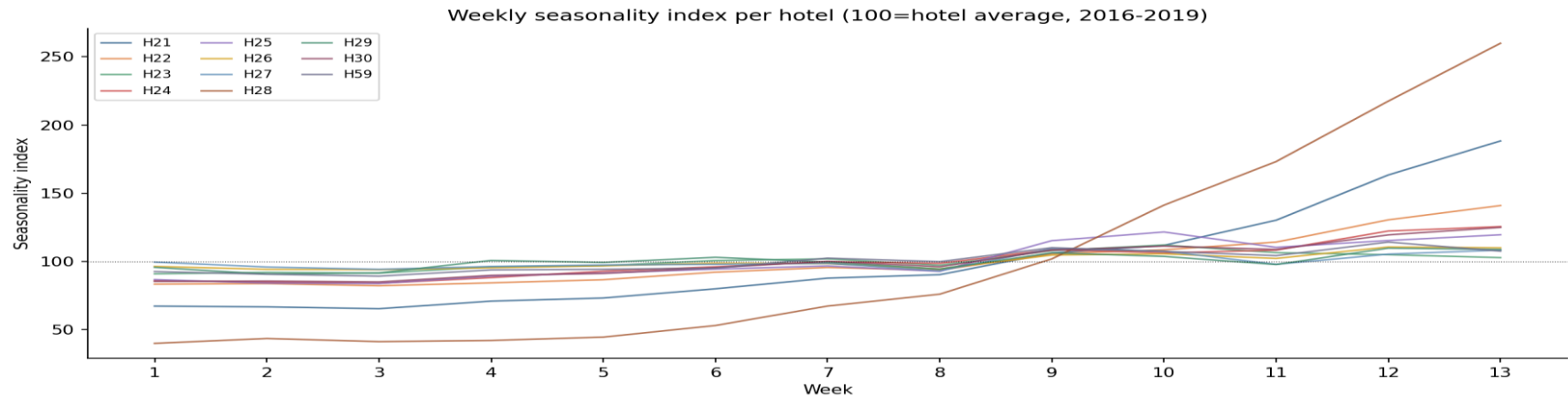
A1: Mean weekly revenue by hotel (2016-2020 average)



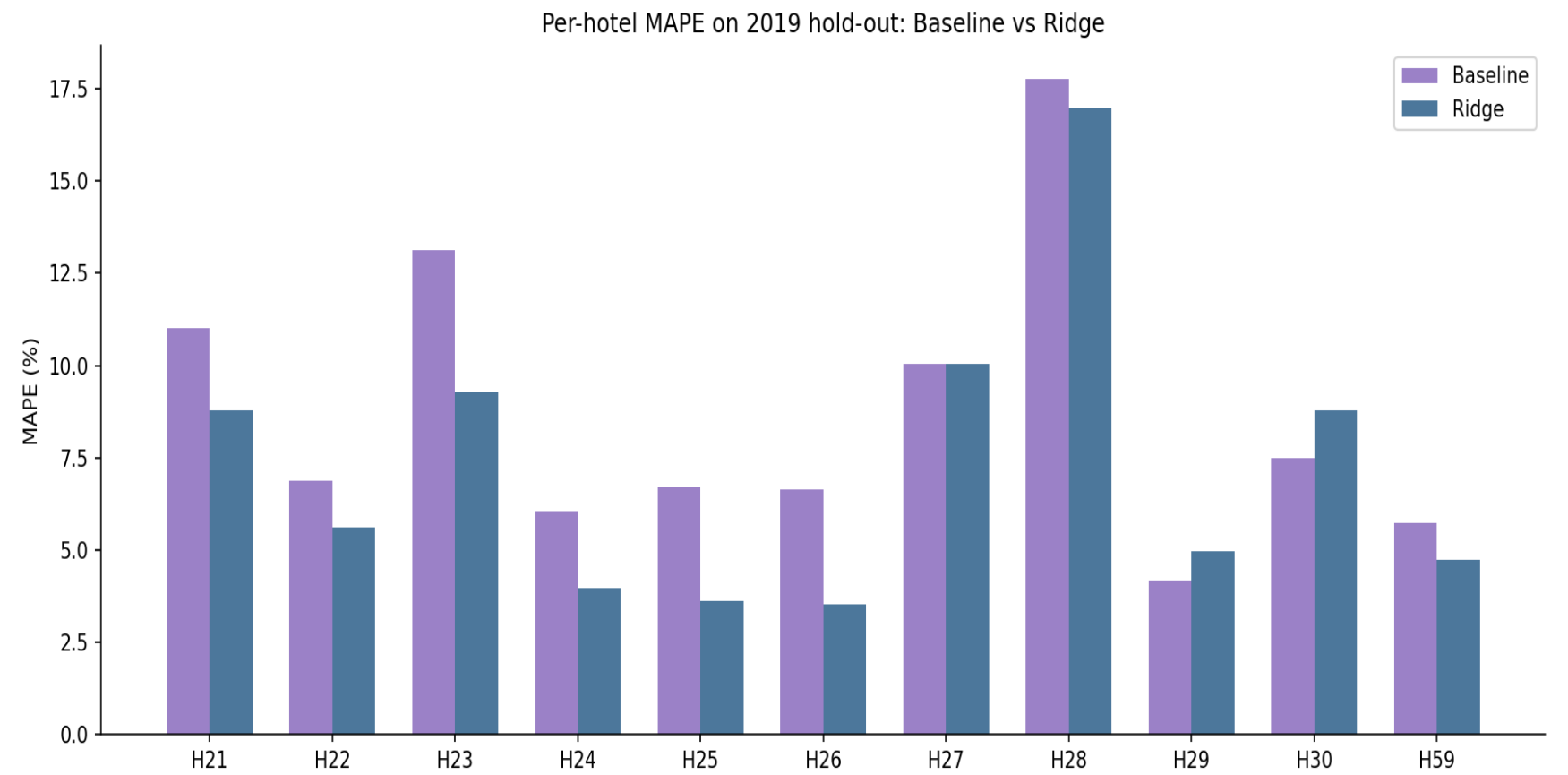
A2: Hotel segmentation — CAGR vs volatility



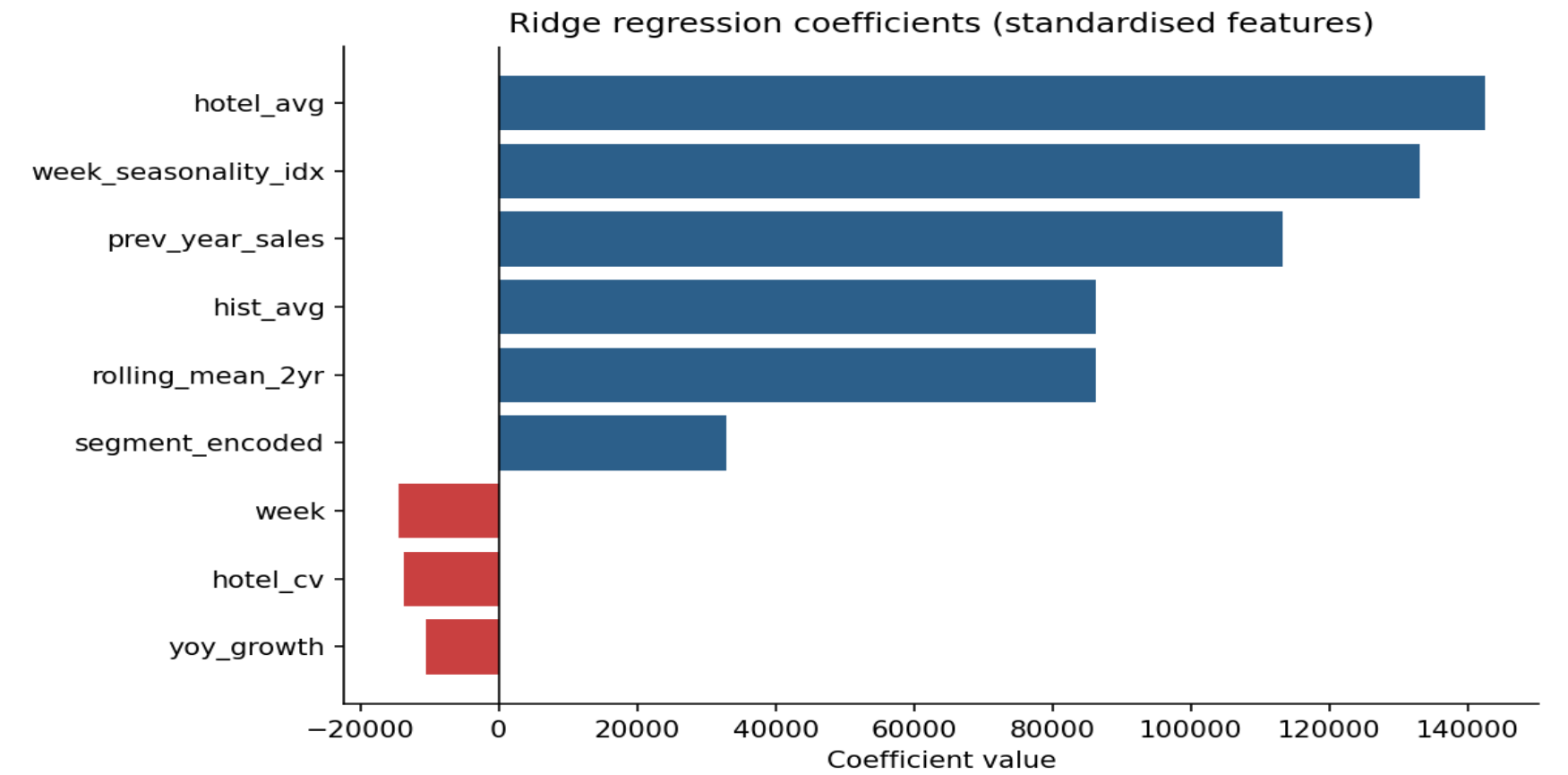
A3: Weekly seasonality index



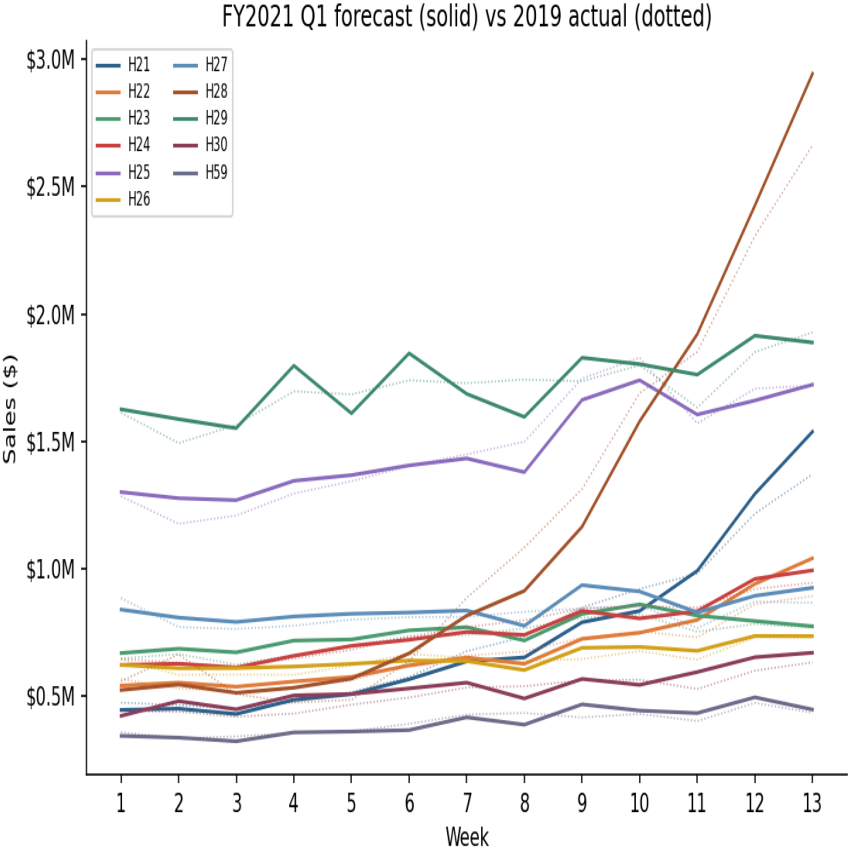
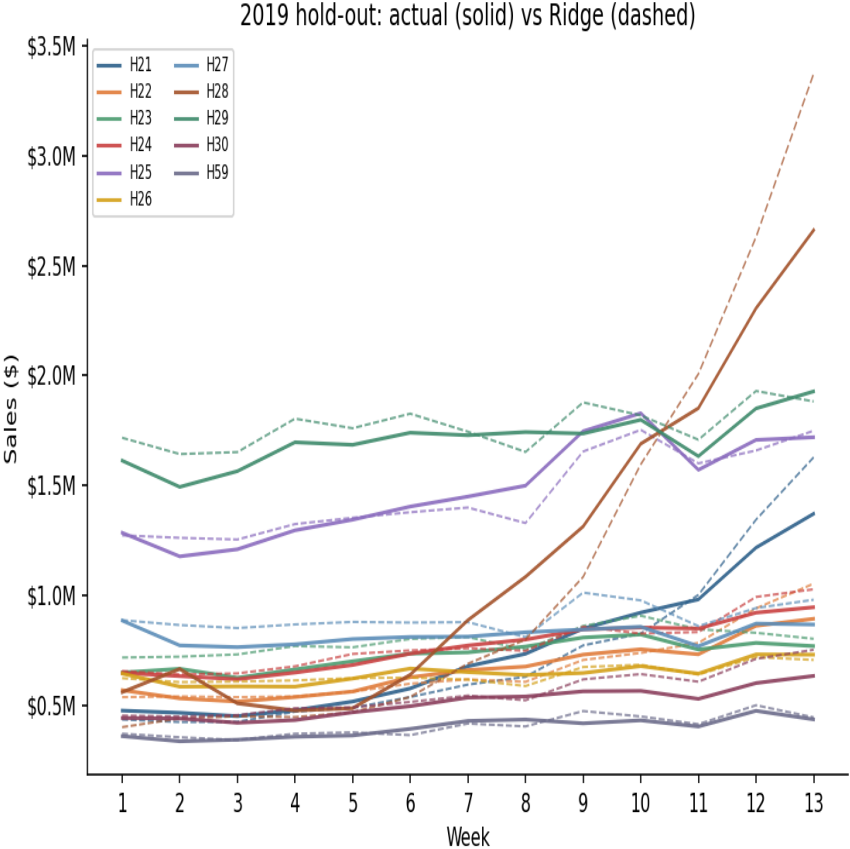
A4: Per-hotel MAPE — Baseline vs Ridge (2019 hold-out)



A5: Ridge regression coefficients



A6: 2019 hold-out fit and FY2021 Q1 forecast



A7: AI Usage Declaration

Tool used

Claude (Anthropic), via the Claude.ai web interface with project knowledge enabled.

Purposes

Coding support

Notebook cells for data loading, feature engineering, Ridge regression pipeline, model evaluation, and PPTX slide generation. All code was reviewed, tested against the actual dataset, and corrected where needed before use.

Brainstorming

Model selection tradeoffs (Ridge vs Random Forest vs LightGBM), copilot mode design, and initial agentic workflow architecture. In each case, pros and cons were presented and the final decision was made independently.

Slide structure and charts

Layout planning for all 8 slides and suggestions for chart types (COVID impact bar chart, segmentation bubble, MAPE comparison). Final selection and all interpretations are my own.

One suggestion identified, challenged, and corrected

The initial agentic workflow design included a Data Collection Agent, proposed validating copilot outputs through a secondary LLM call, and assigned the Forecast Monitoring Agent an implausibly broad set of responsibilities across data collection, action execution, and monitoring.

All three were identified and rejected through multiple rounds of challenge. Data collection was reassigned to Dagster (stable, cheaper, deterministic). The LLM validation step was dropped as architecturally unrealistic for a single agent. The Forecast Monitoring Agent was narrowed to one bounded role: detect deviations, invoke the copilot, and route decisions.

The final agent design reflects independent judgment about operational scope, cost, and realism. It is materially different from the AI's initial proposal.